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ALGORITHM AND DATASET

Loan Approval Prediction System

1. Algorithm

1.1 Step-by-Step Procedure

- **Step 1: Data Collection** — Load the loan application dataset containing applicant details and the corresponding loan status.
- **Step 2: Data Cleaning** — Handle missing values in fields such as credit history, loan amount, and self-employment status through imputation.
- **Step 3: Data Encoding** — Convert categorical variables (gender, marital status, education, property area, etc.) into numerical form using label or one-hot encoding.
- **Step 4: Feature Engineering** — Derive new features such as total household income, debt-to-income ratio, and log-transformed loan amount.
- **Step 5: Feature Scaling** — Normalize or standardize numerical features so that no single feature dominates due to its scale.
- **Step 6: Train-Test Split** — Divide the processed dataset into training and testing subsets using stratified sampling to preserve class balance.
- **Step 7: Handling Class Imbalance** — Apply SMOTE (Synthetic Minority Oversampling Technique) on the training set to balance approved and rejected classes.
- **Step 8: Model Training** — Train classification models (Logistic Regression, Random Forest, XGBoost) on the balanced training data.
- **Step 9: Hyperparameter Tuning** — Use grid search with cross-validation to select the best-performing configuration for each model.
- **Step 10: Model Evaluation** — Evaluate all models on the test set using accuracy, precision, recall, F1-score, and ROC-AUC, and select the best model.
- **Step 11: Deployment** — Serialize the final model and serve it through a Streamlit interface for real-time predictions.

1.2 Logic or Method Used

The system follows a supervised binary classification approach. Each applicant record is treated as an independent observation with a set of input features and a known outcome label (approved or rejected). The model learns the statistical relationship between applicant attributes — such as income, credit history, and loan amount — and the final decision, so that it can generalize this relationship to predict outcomes for new, unseen applicants.

1.3 Model or Technique Applied

Three classification techniques are applied and compared: Logistic Regression, which serves as an interpretable linear baseline; Random Forest, an ensemble of decision trees that captures non-linear relationships and feature interactions; and XGBoost, a gradient boosting technique that builds trees sequentially to correct prior errors, typically yielding the highest accuracy. The best-performing model, selected through cross-validated evaluation, is used for the final deployed system.

1.4 Process Flow of the System

- **Input** — Applicant details entered via form or uploaded as a record
- **Preprocessing** — Cleaning, encoding, scaling, and feature engineering applied
- **Model Inference** — Processed feature vector passed to the trained classifier
- **Prediction** — Model outputs an approval or rejection decision with a probability score
- **Output** — Result displayed to the user through the Streamlit interface

2. Dataset

2.1 Source of Data

The dataset used is the publicly available Loan Prediction Dataset from Kaggle, which is commonly used for benchmarking loan approval classification models. It consists of historical loan application records along with the final approval decision for each applicant.

2.2 Number of Records/Samples

The dataset contains 614 applicant records in the training set, along with a separate 367-record test set (unlabeled, used for leaderboard evaluation on Kaggle).

2.3 Features or Attributes (Columns)

- Gender — Male / Female
- Marital Status — Married / Unmarried
- Dependents — Number of dependents
- Education — Graduate / Not Graduate
- Self-Employed — Yes / No
- Applicant Income — Monthly income of the applicant
- Co-applicant Income — Monthly income of the co-applicant, if any

- Loan Amount — Requested loan amount
- Loan Amount Term — Repayment term of the loan, in months
- Credit History — Whether the applicant meets credit history guidelines
- Property Area — Urban / Semi-Urban / Rural
- Loan Status — Target label indicating approval (Y) or rejection (N)

2.4 Training and Testing Data Split

The dataset is split into 80% for training and 20% for testing, using stratified sampling to preserve the same ratio of approved to rejected applications in both subsets. A portion of the training data is further held out for cross-validation during hyperparameter tuning.

2.5 Data Format or Type

The dataset is provided in CSV (Comma-Separated Values) format, with each row representing one applicant and each column representing a feature or the target label. It contains a mix of categorical (text-based) and numerical (continuous and discrete) data types, requiring encoding and scaling before it can be used by the machine learning models.